GITAU DENNIS NJOROGE

Network Engineer | Software Engineer | Full-Stack Web Developer

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PROFESSIONAL SUMMARY

Versatile and solutions-oriented Network and Software Engineer with two years of hands-on experience designing robust multi-tier network topologies, configuring Layer 2/3 enterprise infrastructure, and developing full-stack web applications. Proficient in Cisco IOS administration, VLAN segmentation, inter-VLAN routing, and dynamic protocols, complemented by strong backend and frontend engineering expertise using Python, Django, ASGI architectures, C++, and Linux environments. Adept at bridging the gap between networking infrastructure and software systems to deliver secure, scalable, and high-performance solutions.

CORE TECHNICAL COMPETENCIES

- **Network Engineering & Protocols:** VLANs (802.1Q), Inter-VLAN Routing (Router-on-a-Stick), Spanning Tree Protocol (STP), 802.1Q Trunks, Static Routing, OSPF, DHCP Pools & Relay (ip helper-address), Subnetting/VLSM, TCP/IP, NAT/PAT.
- **Network Design & Diagnostics:** Cisco Packet Tracer, Cisco IOS CLI configuration, Wireshark, ICMP Diagnostics (ping, traceroute), interface and routing table analysis.
- **Software Engineering & Web Development:** Full-Stack Development, Python (Django, ASGI/Daphne), C++ (Multithreading/Concurrency), R, HTML5, CSS3, JavaScript, RESTful APIs.
- **Databases & Systems Administration:** SQLite, PostgreSQL, Linux/Unix administration (Ubuntu LTS), Bash Shell Scripting, Git/GitHub, Environment Configuration (Venv, Snap, APT).

EDUCATION

Bachelor of Science in Information Technology **Expected Graduation: 2028**

Taita Taveta University – Voi, Taita Taveta County, Kenya *Undergraduate*

- Relevant Coursework: Computer Networks, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Web Applications Architecture.

FEATURED TECHNICAL PROJECTS

Enterprise Multi-Tier Campus & Core WAN Topology **Network Architect**

- Designed and deployed a redundant multi-switch and multi-router enterprise network architecture using Cisco Packet Tracer.
- Segmented department traffic across 12 distinct access switches utilizing 802.1Q VLANs and engineered resilient trunk uplinks to distribution switches (Switch12, Switch3, Switch6, Switch9).
- Implemented inter-VLAN routing across core routers (Router0–Router3) using router subinterfaces with IEEE 802.1Q encapsulation, isolating subnets and eliminating broadcast domain leakage.
- Engineered and deployed localized DHCP service pools directly on edge routing interfaces, eliminating cross-network address leakage and automating host IP provisioning.
- Designed a redundant mesh core WAN interconnecting routers over point-to-point /30 subnets and serial lines, establishing end-to-end multi-path connectivity.

Netpulse Engine **Full-Stack Developer & Backend Architect**

- Developed an asynchronous web application engine utilizing Python, Django, and the Daphne ASGI server.
- Engineered secure user authentication workflows, database-driven signup/login logic, and model migration pipelines.
- Integrated and optimized relational database models using Django ORM with SQLite, resolving database synchronization issues and migration conflicts.
- Built real-time asynchronous request handling routines and streamlined local development workflows on Linux.

TECHNICAL EXPERIENCE

Network & Software Engineering Associate (Independent & Academic Projects) **2024 – Present**

Kenya *Project-Based / Freelance*

- Architected, configured, and tested Layer 2 switching and Layer 3 routing configurations for simulated enterprise networks.
- Diagnosed low-level operational network faults, including IP subnet overlaps on parent router interfaces, trunking native VLAN mismatches, and DHCP relay failures.
- Developed modular backend scripts in Python and high-performance algorithms in C++ incorporating multi-threaded execution.
- Conducted structured statistical data analysis and command automation utilizing R programming and shell utilities.

CERTIFICATIONS & CONTINUOUS LEARNING

- **Cisco Certified Network Associate (CCNA)** – Preparation in progress
- **Advanced Python & Django Web Framework Specialization** – Ongoing